

# MATH 152 — PART 1

Number Systems, Inequalities, Absolute Values, and Point Sets  
Detailed Notes with Tips and Examples

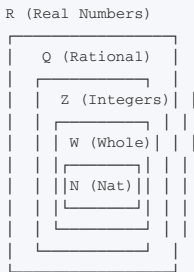
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## 1.0 Introduction to Number Systems

### The Hierarchy of Numbers

Understanding the relationship between different number sets is fundamental to mathematics. Think of it as a Russian nesting doll — each set contains the previous ones.



### Detailed Breakdown

#### Natural Numbers (N) = {1, 2, 3, 4, ...}

##### Key Properties:

- Also called "counting numbers"
- Closed** under addition:  $2 + 3 = 5 \in \mathbb{N}$
- Closed** under multiplication:  $2 \times 3 = 6 \in \mathbb{N}$
- NOT closed** under subtraction:  $2 - 3 = -1 \notin \mathbb{N}$
- NOT closed** under division:  $2 \div 3 = 2/3 \notin \mathbb{N}$

**TIP:** "Closed" means performing the operation on any two numbers in the set always gives a result that's ALSO in the set.

#### Whole Numbers (W) = {0, 1, 2, 3, 4, ...}

##### Key Properties:

- $\mathbb{N} \cup \{0\} = \mathbb{W}$
- Adding zero as a "placeholder"

#### Integers (Z) = {..., -3, -2, -1, 0, 1, 2, 3, ...}

##### Key Properties:

- Closed** under addition:  $(-3) + 5 = 2 \in \mathbb{Z}$
- Closed** under multiplication:  $(-2) \times 4 = -8 \in \mathbb{Z}$
- Closed** under subtraction:  $3 - 5 = -2 \in \mathbb{Z}$
- NOT closed** under division:  $3 \div 2 = 1.5 \notin \mathbb{Z}$

##### Subsets of Z:

- Positive integers:**  $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$
- Negative integers:**  $\mathbb{Z}^- = \{\dots, -3, -2, -1\}$
- Non-negative integers:**  $\{0\} \cup \mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$

**TIP:** Remember:  $\mathbb{Z} = \mathbb{Z}^- \cup \{0\} \cup \mathbb{Z}^+$

#### Rational Numbers (Q) = {a/b : a, b ∈ Z, b ≠ 0}

##### Key Properties:

- Closed** under ALL arithmetic operations (+, −, ×, ÷)
- Exception:** Division by zero is undefined!

Important Notes:

- 1.  $0 \in \mathbb{Q}$  (since  $0 = 0/1$ ,  $b \neq 0$ )
- 2. Every integer is rational ( $a = a/1$ )
- 3. So  $\mathbb{Z} \subset \mathbb{Q}$

Examples of Rational Numbers:  $3/4$  (proper fraction),  $5/2$  (improper fraction),  $0.75$  (terminating decimal),  $0.333...$  (recurring decimal)

🔑KEY INSIGHT: A number is rational if it can be expressed as a fraction of two integers (with denominator  $\neq 0$ ).

Irrational Numbers (I)

Definition: Numbers that CANNOT be expressed in the form  $a/b$  where  $a, b \in \mathbb{Z}$ .

Common Examples:  $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt{5}$ ,  $\sqrt{7}$  (square roots of non-perfect squares);  $\pi \approx 3.14159...$ ;  $e \approx 2.71828...$ ;  $\sqrt{2} + 1$  (sum of rational and irrational is irrational)

Decimal Representation:

| Type                           | Decimal Form             | Example                    |
|--------------------------------|--------------------------|----------------------------|
| Terminating decimal            | Ends after finite digits | $1/2 = 0.5$                |
| Recurring decimal              | Repeats pattern          | $1/3 = 0.333...$           |
| Non-terminating, non-recurring | Never ends, no pattern   | $\sqrt{2} = 1.41421356...$ |

The Proof That  $\sqrt{2}$  is Irrational

This is a CLASSIC proof by contradiction. Memorize the structure — it appears in exams frequently!

STEP-BY-STEP PROOF

Step 1: Assume the opposite  
Assume  $\sqrt{2}$  is rational.  
Then  $\sqrt{2} = a/b$ , where  $a, b \in \mathbb{Z}$ ,  $b \neq 0$ , and  $a/b$  is in lowest terms.

Step 2: Square both sides  
 $2 = a^2/b^2$   
 $\Rightarrow a^2 = 2b^2$

Step 3: Analyze  $a^2$   
 $a^2 = 2b^2 \rightarrow a^2$  is even (multiple of 2)  
If  $a^2$  is even, then  $a$  is even (only even numbers square to even numbers)  
So  $a = 2c$  for some integer  $c$ .

Step 4: Substitute  
 $(2c)^2 = 2b^2$   
 $4c^2 = 2b^2$   
 $b^2 = 2c^2$

Step 5: Analyze  $b^2$   
 $b^2 = 2c^2 \rightarrow b^2$  is even  $\rightarrow b$  is even

Step 6: Contradiction!  
Both  $a$  and  $b$  are even.  
So  $a/b$  can be simplified (divided by 2).  
But we assumed  $a/b$  was in lowest terms.

CONTRADICTION!  
Therefore,  $\sqrt{2}$  is irrational.

💡TIP: This proof uses the following facts: (1) A number is even  $\rightarrow$  its square is even. (2) A square is even  $\rightarrow$  the number is even. (3) If both numerator and denominator are even, the fraction can be simplified.

1.1 Operations with Real Numbers

The Field Axioms

A set that satisfies these rules is called a **FIELD**. The real numbers  $\mathbb{R}$  form a field.

| Rule | Name                     | Mathematical Statement                           |
|------|--------------------------|--------------------------------------------------|
| 1    | Closure (+)              | $a + b \in \mathbb{R}$                           |
| 2    | Closure ( $\times$ )     | $ab \in \mathbb{R}$                              |
| 3    | Commutative (+)          | $a + b = b + a$                                  |
| 4    | Commutative ( $\times$ ) | $ab = ba$                                        |
| 5    | Associative (+)          | $(a+b)+c = a+(b+c)$                              |
| 6    | Associative ( $\times$ ) | $(ab)c = a(bc)$                                  |
| 7    | Distributive             | $a(b+c) = ab + ac$                               |
| 8    | Identity (+)             | $a + 0 = 0 + a = a$                              |
| 9    | Identity ( $\times$ )    | $a \cdot 1 = 1 \cdot a = a$                      |
| 10   | Inverse (+)              | $a + (-a) = 0$                                   |
| 11   | Inverse ( $\times$ )     | $a \cdot a^{-1} = 1 \text{ (} a \neq 0 \text{)}$ |

💡**TIP:** These axioms are the "rules of the game" for algebra. Every algebraic manipulation relies on these fundamental laws.

## 1.2 Inequalities

### Fundamental Theorems

#### Theorem 1: Law of Trichotomy

For any  $a, b \in \mathbb{R}$ , exactly ONE of the following is true:  $a > b$  OR  $a = b$  OR  $a < b$

💡**TIP:** This simply means any two numbers can be compared — one is greater, equal, or less than the other.

#### Theorem 2: Law of Transitivity

If  $a > b$  and  $b > c$ , then  $a > c$   
*Example: If  $5 > 3$  and  $3 > 1$ , then  $5 > 1$*

💡**TIP:** This allows us to "chain" inequalities together.

#### Theorem 3: Addition Property

If  $a > b$ , then  $a + c > b + c$   
*Example: If  $5 > 3$ , then  $5 + 2 > 3 + 2$  ( $7 > 5$ )*

#### Theorem 4: Multiplication by Positive Number

If  $a > b$  and  $c > 0$ , then  $ac > bc$   
*Example: If  $5 > 3$ , then  $5 \times 2 > 3 \times 2$  ( $10 > 6$ )*

#### Theorem 5: Multiplication by Negative Number

If  $a > b$  and  $c < 0$ , then  $ac < bc$

⚠️**CRITICAL WARNING:** When multiplying or dividing an inequality by a **negative** number, you **MUST** flip the inequality sign!

Example:  
 $-2x < 6$   
Divide both sides by  $-2$  (negative!)  
 $x > -3$  ← SIGN FLIPPED!

💡**TIP:** Many students forget this. Always remember: **NEGATIVE  $\times$  SIGN FLIP!**

### Solving Quadratic Inequalities

#### Method 1: Case Analysis (Most Common)

EXAMPLE — Solve  $x^2 - x - 2 > 0$

```
Step 1: Factor
(x + 1)(x - 2) > 0

Step 2: For product > 0, both factors must have same sign
Case 1: Both positive
x + 1 > 0 AND x - 2 > 0
x > -1 AND x > 2
x > 2 (intersection)

Case 2: Both negative
x + 1 < 0 AND x - 2 < 0
x < -1 AND x < 2
x < -1 (intersection)

Step 3: Combine
{x : x < -1} ∪ {x : x > 2}
```

💡TIP: For  $> 0$  (positive product), factors must have SAME sign. For  $< 0$  (negative product), factors must have OPPOSITE signs.

Method 2: Graphical Method

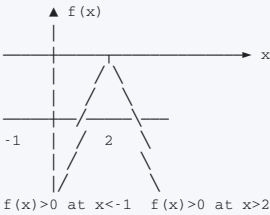
Steps: (1) Factor the expression. (2) Find roots (where expression = 0). (3) Sketch the parabola. (4) Identify regions where  $y > 0$  or  $y < 0$ .

EXAMPLE — Solve  $x^2 - x - 2 > 0$

```
f(x) = (x + 1)(x - 2)
Roots: x = -1, x = 2

The parabola opens upward (coefficient of x² is positive)
Regions:
- x < -1: f(x) > 0 ✓
- -1 < x < 2: f(x) < 0 ✗
- x > 2: f(x) > 0 ✓

Solution: x < -1 or x > 2
```



Method 3: Sign Table Method

Steps: (1) Find roots. (2) Create a table with intervals between roots. (3) Determine sign of each factor in each interval. (4) Determine overall sign.

EXAMPLE — Solve  $x^2 - x - 2 > 0$

$f(x) = (x + 1)(x - 2)$ , Roots:  $x = -1, x = 2$

| Interval    | $x < -1$ | $-1 < x < 2$ | $x > 2$ |
|-------------|----------|--------------|---------|
| $x + 1$     | −        | +            | +       |
| $x - 2$     | −        | −            | +       |
| <b>f(x)</b> | +        | −            | +       |

Solution:  $x < -1$  or  $x > 2$

💡TIP: The sign table method is the most systematic and least prone to errors!

Special Cases

Case: Product  $\geq 0$  or  $\leq 0$

Example:  $x^2 - x - 2 \geq 0$  — Solution includes roots!  
 $\{x : x \leq -1\} \cup \{x : x \geq 2\}$

💡TIP:  $\geq$  includes the roots,  $>$  excludes them.

Case: Negative Leading Coefficient

Example:  $9 - 3x - 2x^2 \leq 0$

Option 1 — Multiply by −1:

$$9 - 3x - 2x^2 \leq 0$$

Multiply by -1 (FLIP sign!)

$$2x^2 + 3x - 9 \geq 0$$

Option 2 — Factor directly:

$$9 - 3x - 2x^2 = (3 - 2x)(3 + x) \leq 0$$

💡**TIP:** For parabolas: Coefficient of  $x^2 > 0 \rightarrow$  "smile" shape  $\cup$  (opens upward). Coefficient of  $x^2 < 0 \rightarrow$  "frown" shape  $\cap$  (opens downward).

Common Mistakes to Avoid

| Mistake                                              | Correct Way                                                 |
|------------------------------------------------------|-------------------------------------------------------------|
| Forgetting to flip sign when multiplying by negative | Always flip when multiplying/dividing by negative           |
| Using AND instead of OR                              | Different regions use OR; overlapping regions use AND       |
| Including roots for strict inequalities              | $>$ and $<$ exclude roots; $\geq$ and $\leq$ include roots  |
| Not checking the sign of a                           | Always consider that a could be negative in $ax^2 + bx + c$ |

### 1.3 Absolute Value

#### Definition

The absolute value of  $x$ , denoted  $|x|$ , is the **distance** of  $x$  from zero on the number line.

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

🔑**KEY INSIGHT:** Absolute value is NEVER negative. It only asks "how far?" not "in which direction?"

**Examples:**  $|3| = 3$  (3 units to the right of 0).  $|-3| = 3$  (3 units to the left of 0).  $|0| = 0$  (0 units from 0).

#### Properties of Absolute Value

| Property            | Formula                         | Explanation                                    |
|---------------------|---------------------------------|------------------------------------------------|
| Non-negativity      | $ x  \geq 0$                    | Absolute value is never negative               |
| Product             | $ xy  =  x  y $                 | Absolute value distributes over multiplication |
| Triangle Inequality | $ x + y  \leq  x  +  y $        | Sum of absolutes is at least absolute of sum   |
| Reverse Triangle    | $ x - y  \geq   x  -  y  $      | Difference inequality                          |
| Zero                | $ x  = 0 \Leftrightarrow x = 0$ | Only zero has absolute value zero              |

#### Proving the Properties

##### PROOF OF $|xy| = |x||y|$

Case 1:  $x > 0, y > 0$   
 $xy > 0 \rightarrow |xy| = xy = |x||y|$

Case 2:  $x < 0, y > 0$   
 $xy < 0 \rightarrow |xy| = -(xy) = (-x)y = |x||y|$

Case 3:  $x < 0, y < 0$   
 $xy > 0 \rightarrow |xy| = xy = (-x)(-y) = |x||y|$

Case 4:  $x = 0$  or  $y = 0$   
 $|xy| = 0 = |x||y|$

Therefore,  $|xy| = |x||y|$  for all real  $x,y$ .

##### PROOF OF TRIANGLE INEQUALITY: $|x + y| \leq |x| + |y|$

Case 1:  $x \geq 0, y \geq 0$   
 $x + y \geq 0 \rightarrow |x + y| = x + y = |x| + |y|$

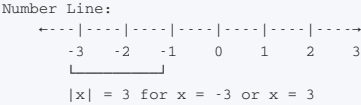
Case 2:  $x < 0, y < 0$   
 $x + y < 0 \rightarrow |x + y| = -(x + y) = (-x) + (-y) = |x| + |y|$

Case 3:  $x \geq 0, y < 0$  (and  $|x| > |y|$ )  
 $x + y > 0 \rightarrow |x + y| = x + y = |x| - |y| < |x| + |y|$

Case 4:  $x \geq 0, y < 0$  (and  $|x| < |y|$ )  
 $x + y < 0 \rightarrow |x + y| = -(x + y) = |y| - |x| < |x| + |y|$

Therefore,  $|x + y| \leq |x| + |y|$  for all real  $x,y$ .

#### Geometric Interpretation



#### Distance Between Points:

$|x - y|$  = distance between  $x$  and  $y$  on the number line  
 $|x - y| = |y - x|$  (symmetric)

**Example:**  $|5 - 2| = 3$  (distance from 2 to 5).  $|2 - 5| = 3$  (distance from 5 to 2).

#### Extended Properties

**Product of  $n$  numbers:**  $|x_1x_2...x_n| = |x_1||x_2|...|x_n|$

Sum of n numbers:  $|x_1 + x_2 + \dots + x_n| \leq |x_1| + |x_2| + \dots + |x_n|$

Simplifying Absolute Value Expressions

| Expression    | Simplified | Explanation                |
|---------------|------------|----------------------------|
| $ 0 - 6 $     | 6          | $ -6  = 6$                 |
| $ 5 - 2 $     | 3          | $ 3  = 3$                  |
| $ 2 - 5 $     | 3          | $ -3  = 3$                 |
| $ 0(-4) $     | 0          | $ 0  = 0$                  |
| $ 2 + 3(-4) $ | 10         | $ 2 + (-12)  =  -10  = 10$ |
| $- -4 $       | -4         | $-(4) = -4$                |
| $- (-3)^2 $   | -9         | $- 9  = -9$                |

💡**TIP:** Always evaluate inside the absolute value bars first!

# 1.4 Point Sets

## Definition

A **point set** is any set whose members are real numbers. All elements can be located on the real number line.

**Examples:** {0, 1, 2, 3, 4, 5} (finite point set); {x : -1 < x < 0} (infinite point set — interval)

## Intervals

### Bounded Intervals

| Type      | Notation | Set Notation    | Example                |
|-----------|----------|-----------------|------------------------|
| Closed    | [a,b]    | {x : a ≤ x ≤ b} | [0,1] includes 0 and 1 |
| Open      | (a,b)    | {x : a < x < b} | (0,1) excludes 0 and 1 |
| Half-open | (a,b]    | {x : a < x ≤ b} | (0,1] includes 1 only  |
| Half-open | [a,b)    | {x : a ≤ x < b} | [0,1) includes 0 only  |

Number Line Representation:

[a,b]

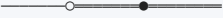
(a,b)

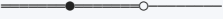
(a,b]

[a,b)









(filled circles = included)

(empty circles = excluded)

### Unbounded Intervals

| Type         | Notation            | Set Notation |
|--------------|---------------------|--------------|
| Open left    | $(-\infty, b)$      | {x : x < b}  |
| Closed left  | $(-\infty, b]$      | {x : x ≤ b}  |
| Open right   | $(a, \infty)$       | {x : x > a}  |
| Closed right | $[a, \infty)$       | {x : x ≥ a}  |
| All real     | $(-\infty, \infty)$ | R            |

**Important:** ∞ and −∞ are NOT real numbers — they're just notation indicating "unbounded."

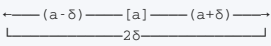
💡**TIP:** ∞ always gets parentheses (never a bracket) because it's not a number that can be included.

## Neighbourhoods

### δ-Neighbourhood of a point a

$$N_\delta(a) = \{x : |x - a| < \delta\} = (a - \delta, a + \delta)$$

Geometric Interpretation:



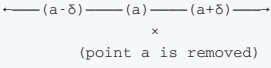
Center = a, Length = 2δ

**Example:** δ-neighbourhood of 2 with δ = 0.5:  $N_{0.5}(2) = (1.5, 2.5)$

### Deleted δ-Neighbourhood

$$N_\delta^*(a) = \{x : 0 < |x - a| < \delta\} = (a - \delta, a) \cup (a, a + \delta)$$

**Difference:** Excludes the point a itself.

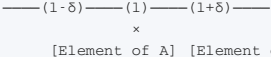


## Limit Points (Cluster Points)

### Definition

A number l is a **limit point** of set A if EVERY deleted δ-neighbourhood of l contains elements of A.

In other words: For every δ > 0, ∃ x ∈ A such that 0 < |x - l| < δ.



### Key Characteristics:

- δ can be as small as we please
- Every δ-neighbourhood contains infinitely many elements of A
- Limit points don't have to be in the set itself



Examples of Limit Points

Example 1: A = (0, 1)

Limit points: 0 and 1 (and all points in between)

Why is 0 a limit point? Every  $\delta$ -neighbourhood of 0, i.e.  $(-\delta, \delta)$ , will contain elements of (0,1) for any  $\delta > 0$ . But 0 itself is NOT in the set (it's open).

Example 2: Finite Set {1, 2, 3, 4, 5}

Limit points: NONE

Why? For any suggested number l, we can choose  $\delta$  small enough that the deleted  $\delta$ -neighbourhood contains no elements of the set.

Example 3: N = {1, 2, 3, 4, ...}

Limit points: NONE

Why? The natural numbers "run away" to infinity. For any proposed limit, we can find a small enough neighbourhood with no natural numbers.

Example 4: A = {1, 1/2, 1/3, 1/4, ...}

Limit point: 0

Why? Elements get arbitrarily close to 0 but never reach it.

Example 5: A = Q (Rational Numbers)

Limit points: ALL real numbers

Why? Every interval contains rational numbers.

Closed Sets

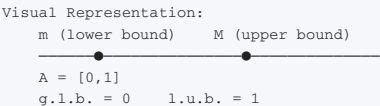
Definition: A set is **closed** if it contains ALL its limit points.

| Set                           | Closed? | Why?                           |
|-------------------------------|---------|--------------------------------|
| [0,1]                         | Yes     | Contains 0 and 1               |
| (0,1)                         | No      | Does NOT contain 0 and 1       |
| {1, 1/2, 1/3, ...} $\cup$ {0} | Yes     | Contains limit point 0         |
| {1, 1/2, 1/3, ...}            | No      | Does NOT contain limit point 0 |

Bounds

Definitions

| Term                          | Definition                                        | Example for A = [0,1]   |
|-------------------------------|---------------------------------------------------|-------------------------|
| Bounded above                 | $\exists M$ such that $x \leq M, \forall x \in A$ | $M = 2, 3, 5$ , etc.    |
| Upper bound                   | Any M that bounds A above                         | 2, 3, 5, ...            |
| Least upper bound (l.u.b.)    | Smallest upper bound                              | 1                       |
| Bounded below                 | $\exists m$ such that $x \geq m, \forall x \in A$ | $m = -1, -2, -5$ , etc. |
| Lower bound                   | Any m that bounds A below                         | -1, -2, -5, ...         |
| Greatest lower bound (g.l.b.) | Largest lower bound                               | 0                       |



Examples of Bounds

| Set                | Upper Bounds | l.u.b.         | Lower Bounds   | g.l.b. | Bounded?               |
|--------------------|--------------|----------------|----------------|--------|------------------------|
| [0,1]              | 1, 2, 3, ... | 1              | 0, -1, -2, ... | 0      | Yes                    |
| (0,1)              | 1, 2, 3, ... | 1              | 0, -1, -2, ... | 0      | Yes*                   |
| {1, 1/2, 1/3, ...} | 1, 2, 3, ... | 1              | 0, -1, -2, ... | 0      | Yes                    |
| N = {1, 2, 3, ...} | NONE         | Does not exist | 1, 0, -1, ...  | 1      | No (not bounded above) |

💡TIP: \*For (0,1): l.u.b. and g.l.b. may NOT be in the set!

Bolzano-Weierstrass Theorem

🔑KEY INSIGHT: Every bounded infinite set has at least one limit point.

**What This Means:** If a set is bounded and infinite, it must "accumulate" somewhere. This is a fundamental theorem in real analysis.

| Set                               | Bounded?                  | Infinite? | Limit Point?                   |
|-----------------------------------|---------------------------|-----------|--------------------------------|
| $A = \{1, 1/2, 1/3, 1/4, \dots\}$ | Yes ( $0 \leq x \leq 1$ ) | Yes       | $0$ ✓ (satisfies theorem)      |
| $A = \{1, 2, 3, 4, \dots\}$       | No (not bounded above)    | Yes       | None ✓ (theorem doesn't apply) |
| $A = (0,1)$                       | Yes                       | Yes       | $[0,1]$ (infinitely many) ✓    |

Practice Problems and Solutions

PROBLEM 1

Find the truth set of  $x^2 - 7x + 10 \geq 0$

Solution:

$$x^2 - 7x + 10 = (x - 2)(x - 5)$$

Roots:  $x = 2, x = 5$

| Interval | $x < 2$ | $2 < x < 5$ | $x > 5$ |
|----------|---------|-------------|---------|
| $x - 2$  | -       | +           | +       |
| $x - 5$  | -       | -           | +       |
| $f(x)$   | +       | -           | +       |

Since we want  $f(x) \geq 0$ :  
 $\{x : x \leq 2\} \cup \{x : x \geq 5\}$

PROBLEM 2

Solve  $6x^2 - x - 35 \leq 0$

Solution:

$$6x^2 - x - 35 = (3x + 7)(2x - 5)$$

Roots:  $x = -7/3, x = 5/2$

| Interval | $x < -7/3$ | $-7/3 < x < 5/2$ | $x > 5/2$ |
|----------|------------|------------------|-----------|
| $3x + 7$ | -          | +                | +         |
| $2x - 5$ | -          | -                | +         |
| $f(x)$   | +          | -                | +         |

Since we want  $f(x) \leq 0$ :  
 $\{x : -7/3 \leq x \leq 5/2\}$

PROBLEM 3

Solve  $(1 - 2x)(x + 2) < 0$

Solution:

Roots:  $1 - 2x = 0 \rightarrow x = 1/2$   
 $x + 2 = 0 \rightarrow x = -2$

| Interval | $x < -2$ | $-2 < x < 1/2$ | $x > 1/2$ |
|----------|----------|----------------|-----------|
| $1 - 2x$ | +        | +              | -         |
| $x + 2$  | -        | +              | +         |
| $f(x)$   | -        | +              | -         |

Since we want  $f(x) < 0$ :  
 $\{x : x < -2\} \cup \{x : x > 1/2\}$

PROBLEM 4

Find the truth set of  $35 - 2x - x^2 \geq 0$

Solution (Method 1 — Multiply by  $-1$ ):

$$35 - 2x - x^2 \geq 0$$

Multiply by  $-1$  (FLIP sign):  
$$x^2 + 2x - 35 \leq 0$$
$$(x + 7)(x - 5) \leq 0$$

Roots:  $x = -7, x = 5$   
For  $\leq 0$ :  $-7 \leq x \leq 5$

Solution (Method 2 — Work directly):

$$35 - 2x - x^2 = -(x^2 + 2x - 35) = -(x + 7)(x - 5)$$

For  $\geq 0$ : need  $(x + 7)(x - 5) \leq 0$   
 $-7 \leq x \leq 5$

PROBLEM 5

Find the truth set of  $3x^2 - x + 2 > 0$

Solution:

Discriminant:  $\Delta = b^2 - 4ac = (-1)^2 - 4(3)(2) = 1 - 24 = -23$   
Since  $\Delta < 0$ , the parabola never crosses x-axis.  
 $a = 3 > 0 \rightarrow$  parabola opens upward  $\rightarrow$  always positive.

Truth set: All real numbers  $\mathbb{R}$

✂️TIP: If discriminant is negative and  $a > 0$ , the quadratic is always positive!

PROBLEM 6

Find the truth set of  $(3 - 2x)(4 - 2x) \leq 0$

Solution:

Roots:  $3 - 2x = 0 \rightarrow x = 3/2$   
 $4 - 2x = 0 \rightarrow x = 2$

| Interval | $x < 3/2$ | $3/2 < x < 2$ | $x > 2$ |
|----------|-----------|---------------|---------|
| $3 - 2x$ | +         | -             | -       |
| $4 - 2x$ | +         | +             | -       |
| $f(x)$   | +         | -             | +       |

Since we want  $f(x) \leq 0$ :  
 $\{x : 3/2 \leq x \leq 2\}$

PROBLEM 7

Find the cluster point(s) of  $A = \{1, 1/2, 1, 1/3, 1, 1/4, 1, 1/5, \dots\}$

Solution:

The sequence is:  $1, 1/2, 1, 1/3, 1, 1/4, 1, 1/5, \dots$   
Elements: All numbers of the form  $1/n$  ( $n \geq 2$ ) and the number 1.

Limit points:  
- 1 is a limit point (repeated infinitely)  
- 0 is a limit point ( $1/n \rightarrow 0$ )

So the cluster points are:  $\{0, 1\}$

PROBLEM 8

Does  $T = \{1, 2, 3, 4, 5\}$  have a cluster point?

Solution:

No!  $T$  is a finite set.  
Finite sets have NO cluster points.

For any proposed number 1, we can choose  $\delta$  small enough that the deleted  $\delta$ -neighbourhood contains no elements of  $T$ .

PROBLEM 9

Find the limit point(s) of  $a_n = 1 + (-1)^n + 1/n$

Solution:

$a_n = 1 + (-1)^n + 1/n$

When  $n$  is even:  $(-1)^n = 1$   
 $a_n = 1 + 1 + 1/n = 2 + 1/n \rightarrow 2$

When  $n$  is odd:  $(-1)^n = -1$   
 $a_n = 1 - 1 + 1/n = 1/n \rightarrow 0$

So the sequence has two subsequences converging to different limits.

Limit points:  $\{0, 2\}$

**PROBLEM 10**

**Find the limit point(s) of  $A = \{1/2, 1/3, 2/3, 1/4, 2/4, 3/4, 1/5, 2/5, 3/5, 4/5, \dots\}$**

**Solution:**

$A$  = all fractions between 0 and 1 of the form  $k/n$  where  $1 \leq k < n$ .  
This set is actually ALL rational numbers in  $(0,1)$ .  
Every rational number appears in this sequence.

Limit points of  $Q \cap (0,1) = [0,1]$

So the cluster points are all numbers in  $[0,1]$ .

# Common Mistakes and How to Avoid Them

## Mistake 1: Forgetting to Flip the Inequality Sign

| Wrong                          | Correct                                                                        |
|--------------------------------|--------------------------------------------------------------------------------|
| $-3x < 6 \Rightarrow x < -2$ ✗ | $-3x < 6 \Rightarrow x > -2$ ✓ (dividing by negative $\rightarrow$ flip sign!) |

💡**TIP:** Always remember NEGATIVE = FLIP!

## Mistake 2: Confusing AND with OR

**Example:**  $(x + 1)(x - 2) > 0$

| Wrong                                                                         | Correct                                                     |
|-------------------------------------------------------------------------------|-------------------------------------------------------------|
| $x > -1$ AND $x > 2 \rightarrow x > 2$ (only gives one part of the solution!) | $x < -1$ OR $x > 2$ (both positive and both negative cases) |

💡**TIP:** For products  $> 0$ , use OR. For products  $< 0$ , use AND (between the roots).

## Mistake 3: Including/Excluding Roots Incorrectly

**For  $>$  or  $<$ :** Exclude roots (open circles). **For  $\geq$  or  $\leq$ :** Include roots (closed circles).

|                                                      |
|------------------------------------------------------|
| $x^2 - 4 > 0 \rightarrow x < -2$ or $x > 2$          |
| $x^2 - 4 \geq 0 \rightarrow x \leq -2$ or $x \geq 2$ |

## Mistake 4: Not Checking if the Quadratic is Always Positive/Negative

⚠️**CRITICAL WARNING:** Always check the discriminant! If  $\Delta < 0$  and  $a > 0 \rightarrow$  always positive. If  $\Delta < 0$  and  $a < 0 \rightarrow$  always negative.

## Mistake 5: Confusing Cluster Points with Elements of the Set

A limit point doesn't have to be in the set!

**Example:**  $(0,1)$  has limit points 0 and 1, even though  $0, 1 \notin (0,1)$

# Summary of Key Formulas

## Number Systems

$$\begin{aligned} \mathbb{N} &\subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \\ \mathbb{Q} \cup \mathbb{I} &= \mathbb{R} \end{aligned}$$

## Inequalities

$$\begin{aligned} \text{If } a > b \text{ and } c > 0 &\rightarrow ac > bc \\ \text{If } a > b \text{ and } c < 0 &\rightarrow ac < bc \end{aligned}$$

## Absolute Value

$$\begin{aligned} |x| &= \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases} \\ |xy| &= |x||y| \\ |x + y| &\leq |x| + |y| \\ |x - y| &\geq ||x| - |y|| \end{aligned}$$

## Intervals

$$\begin{aligned} [a,b] &= \{x : a \leq x \leq b\} \\ (a,b) &= \{x : a < x < b\} \\ [a,b) &= \{x : a \leq x < b\} \\ (a,b] &= \{x : a < x \leq b\} \end{aligned}$$

## Neighbourhoods

$$\begin{aligned} N_\delta(a) &= \{x : |x - a| < \delta\} \\ N_\delta^*(a) &= \{x : 0 < |x - a| < \delta\} \end{aligned}$$